

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Fairness in Machine Learning Models for Healthcare and Education
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Chosen major:	Data Analytics
Supervisor	Tai Tan Mai
Date of Submission	21 Nov 2025

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

This project examines fairness in machine learning for two critical domains: predicting depression risk and student performance. It focuses on whether models produce unequal outcomes across protected attributes such as gender, race, and income. Given the high social impact of healthcare and education, such biases can lead to harm and unequal opportunities. Using demographic, lifestyle, health, and learning-behavior data from two public datasets as use cases, the project investigates how fairness issues arise and how cross-domain mitigation strategies can address them.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2021. A Survey on Bias and Fairness in Machine Learning. ACM Comput. Surv. 54, 6, Article 115 (July 2022), 35 pages. <https://doi.org/10.1145/3457607>
2. Cheong, J., Kuzucu, S., Kalkan, S., & Gunes, H. (2023). Towards Gender Fairness for Mental Health Prediction. International Joint Conferences on Artificial Intelligence Organization. <https://doi.org/10.17863/CAM.96646>

3. Dang, V.N., Cascarano, A., Mulder, R.H. *et al.* Fairness and bias correction in machine learning for depression prediction across four study populations. *Sci Rep* 14, 7848 (2024). <https://doi.org/10.1038/s41598-024-58427-7>
4. Kesgin, K., Kiraz, S., Kosunalp, S., & Stoycheva, B. (2025). Beyond Performance: Explaining and Ensuring Fairness in Student Academic Performance Prediction with Machine Learning. *Applied Sciences*, 15(15), 8409. <https://doi.org/10.3390/app15158409>
5. Zifeng Liu, Xinyue Jiao, Chenglu Li, & Wanli Xing. (2024). Fair Prediction of Students' Summative Performance Changes Using Online Learning Behavior Data. *Proceedings of the 17th International Conference on Educational Data Mining*, 686--691. <https://doi.org/10.5281/zenodo.12729918>.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Mehrabi et al. (2021) provide a survey of bias and fairness in machine learning, offering the theoretical foundation for this project. They review bias types, fairness definitions, and common evaluation and mitigation strategies, guiding the selection of fairness metrics and the design of our fairness pipeline.

My proposal builds on research showing that machine learning models can produce biased predictions for certain demographic groups. In education, Kesgin et al. (2025) demonstrate that student performance models often overlook fairness issues and highlight the importance of techniques like explainability and fairness-aware modelling. Similarly, Liu et al. (2024) show that even behaviour-based predictions of student performance can lead to unequal outcomes, reinforcing the need for fairness metrics and bias detection.

In the Healthcare domain, Cheong et al. (2023) found that mental health prediction models can exhibit gender bias. They compared different mitigation methods and analyzed bias sources. Their findings highlight the need to evaluate protected attributes in depression risk prediction. In addition, Dang et al. (2024) demonstrate that the same model can exhibit varying fairness behaviors across four study populations in depression prediction tasks. Their findings provide valuable insights for comparing bias across different datasets and domains.

In this duo project, we aim to create a **general, model-agnostic fairness pipeline** that can be applied across different datasets. I will apply it to student performance prediction, while my teammate will work on depression risk classification. Using methods such as reweighing and equalized odds post-processing, we will evaluate whether bias can be reduced without significantly affecting accuracy. This cross-domain approach shows that fairness issues appear in multiple contexts and demonstrates a practical way to make machine learning models more equitable.

What are the research questions that you will attempt to answer? (200 words)

- How do machine learning models used to predict student performance and depression risk show biased or unequal outcomes across different demographic groups?

- Which fairness metrics can reliably detect and measure bias in machine learning models, regardless of the dataset or domain?
- What model-agnostic fairness techniques can reduce bias in machine learning predictions while maintaining reasonable accuracy across different datasets?

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?

We will use Python as our primary development environment. Tools include Jupyter Notebook, numpy, pandas, matplotlib, seaborn, scikit-learn, XGBoost, fairlearn and AIF360.

- What coding/development will you do?

- Loading multiple dataset files and merging them together.
- Data cleaning, preprocessing, and feature engineering.
- Constructing PHQ-9 scores and creating depression labels.
- Training machine learning models. (Logistic Regression, Random Forest, or XGBoost)
- Evaluating model performance. (Accuracy, Recall, Precision, F1-score, etc.)
- Conducting fairness analysis using Fairlearn / AIF360. (equalized odds, etc.)
- Applying bias-mitigation techniques if unfairness is detected.
- Visualising results, including model performance, fairness metrics.

- What data will be used for your investigations?

We will be making use of 2 different datasets to examine our approach.

- Data Source:
 - <https://wwwn.cdc.gov/nchs/nhanes/default.aspx>
 - <https://analyse.kmi.open.ac.uk/open-dataset>
- The first data source uses publicly available data from the U.S. National Health and Nutrition Examination Survey (NHANES) August 2021–August 2023 cycle. The survey covers multiple categories of information including demographics, mental health, lifestyle behaviors, which is suitable for mental health prediction research.
- **Main datasets (1):**
 - DEMO_L: Demographics Data
 - DPQ_L (target variable): Mental Health - Depression Screener
 - This questionnaire includes the 9 questions from the PHQ-9 questionnaire. A total score of 10 or higher indicates a risk of depression.
 - ALQ_L: Alcohol Use
 - SMQ_L: Smoking - Cigarette Use
 - DIQ_L: Diabetes
 - INQ_L: Income
 - BMX_L: Body measures
 - OCQ_L: Occupation

- This is the Open University Learning Analytics Dataset (OULAD), one of the largest publicly available datasets in the education domain.
It contains more than 30,000 students and millions of online learning interactions, making it ideal for fairness evaluation.
- **Main datasets (2):**
 - studentInfo.csv – demographic features
 - studentAssessment.csv – student test scores
 - studentVle.csv – millions of online activity logs
 - courses.csv – course metadata
 - assessments.csv – list of assessments
 - studentRegistration.csv – enrollment & withdrawal information

- Is this data currently available, if not, where will it come from?

- Yes

- What experiments do you expect to run?

- Build machine learning models for predicting tasks.
- Evaluate model differences across various populations.
- Experiment with bias mitigation strategies and examine their impact on fairness and performance.

- What output do you expect to gather?

- Performance Metrics: Accuracy, Recall, Precision, F1-score, confusion matrix
- Fairness Metrics: equalized odds, equal opportunity, and demographic parity

- How will the results be evaluated?

Evaluate model performance through performance metrics, compare differences between groups using fairness metrics, and contrast performance and fairness before and after bias mitigation to determine whether model performance has been lost and model fairness has been improved.